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Artificial intelligence readiness in management education: Scale development and validation

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ABSTRACT

Given the proliferation of artificial intelligence (AI) technology, there is a need for management education academics and practitioners to develop an effective and reliable instrument to assess individual readiness to adapt and embrace AI technology/products. The purpose of this study is to develop and validate an artificial intelligence readiness scale (AIRS) with good psychometric properties. The procedures used in conceptualizing the AIRS construct, generating items, collecting data, and validating the multiple-item scale are described. This study provides evidence of reliability, content validity, criterion-related validity, convergent validity, discriminant validity, and nomological validity by analyzing data from a sample of 174 respondents from different industries. Based on the analysis results, a 13-item AIRS with four dimensions (i.e., self-efficacy and cognition, optimism, collaborative learning and working, and comfort) is developed and validated. A tentative norm of the AIRS is also presented. The proposed AIRS provides several important theoretical and practical implications for developing AI-related management education theories and promoting the usage of AI-based products/educational technology.

1. Introduction

Artificial intelligence (AI) has attracted increasing research attention due to its pivotal role in driving innovation across various technological fields and its widespread applications in social and economic environments (Buiten, 2019; Flavián et al., 2022; Lee & Kwon, 2024; Li et al., 2024). As AI technologies become increasingly integrated into everyday life, understanding the readiness of individuals or organizations to adopt and use AI has become crucial. For example, Flavián et al. (2022) examined how technology readiness and service awareness affect consumers' willingness to use analytical AI-based investment services, finding that individuals with higher levels of technological optimism and lower perceived insecurity were more inclined to use robo-advisors. At the same time, some individuals experience anxiety regarding AI's potential impact—particularly as AI has approached and even surpassed human performance in certain tasks. Notably, AI has consistently defeated world champion Go players (Silver et al., 2017) and top-ranked players in the online strategy game AlphaStar (Q.-Y. Yin, Yang, et al., 2023). These breakthroughs underscore not only the rapid development of AI technologies but also highlight the critical role of AI readiness in terms of individuals' ability to adapt to and adopt

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such technologies.

Although prior studies have developed specialized AI readiness scales in the healthcare sector (Karaca et al., 2021), there remains a need to establish a more comprehensive tool to assess individual AI readiness for several reasons. First, research on AI readiness in other applied contexts, such as education, finance, and the hospitality industry, remains relatively limited (Alam et al., 2025; Damerji & Salimi, 2021; Wang et al., 2023). Second, much of the existing literature has focused on AI readiness at the organizational level (Cooper, 2025; Jöhnk et al., 2021), while relatively little attention has been paid to AI readiness at the individual level (Damerji & Salimi, 2021). Third, there is currently no unified standard for measuring AI readiness (Luo & Zou, 2025). For example, Ramazanoglu and Akın (2025) defined AI readiness as the cognitive, emotional, and behavioral state of an individual or organization that facilitates the adoption and effective use of AI technologies to achieve specific goals. This definition includes aspects such as technological self-efficacy, interpersonal interaction, and ethical awareness. In contrast, Karaca et al. (2021) defined AI readiness in the healthcare context as healthcare providers' preparedness in terms of knowledge, skills, and attitudes, structured around four core dimensions: cognition, capability, vision, and ethics.

In the healthcare context, Ozbek Guven et al. (2024) analyzed data from 542 medical students using an artificial intelligence anxiety scale (AIAS), demographic information, and the Medical Artificial Intelligence Readiness Scale (MAIRS). Their results show that the students had on average a moderate level of AI readiness and relatively high level of anxiety in turn, they identified a negative relationship between AIR and AIA. Researchers further explored how AI-based collaboration models, such as an AI platform for cataract management in clinical settings, could improve diagnostic efficiency and optimize healthcare resource allocation.

In the educational field, the integration of AI has sparked critical reflection among educators and researchers. Hrastinski et al. (2019) observed that while AI is increasingly applied in educational settings, educators often lack a deep understanding of AI's practical applications and associated risks. Wang et al. (2023) examined AI readiness in education, conceptualizing its impact across cognitive, capability, vision, and ethical dimensions. Their findings reveal that teachers with higher AI readiness perceived fewer threats from AI and demonstrated greater job satisfaction and innovation capacity, underscoring AI's positive impact on educational outcomes.

Despite these preliminary efforts, research on AI readiness in education remains fragmented and underdeveloped, underscoring the need for deeper investigation (Shahid et al., 2024). As AI continues to accelerate its integration into teaching, learning, and assessment processes, this gap is particularly critical, since varying levels of readiness may directly affect teachers' instructional effectiveness, students' engagement, and overall educational outcomes. For instance, Yue et al. (2024) found that teachers' AI readiness and attitudes are closely linked not only to their teaching effectiveness but also to students' learning experiences and perceptions of the overall quality of AI-driven education. Moreover, unlike the healthcare sector, where validated instruments such as the MAIRS already exist, the educational domain still lacks a standardized framework for assessing AI readiness at the individual level. Addressing this gap is crucial not only for advancing theoretical understanding but also for providing practical guidance to policymakers, educational institutions, and teachers as they navigate AI-driven transformations in education.

In summary, the purpose of this study is to develop and validate a comprehensive scale to assess individual AI readiness. This will provide reliable evaluation criteria and guidelines for organizations and the public in terms of introducing and promoting AI technologies and products. Given the limitations of current self-report tools, this study investigates the psychological aspects related to individual readiness to adopt AI technologies/products. It defines several dimensions conceptually and operationally to assess the nature and scope of AI readiness. The following process was used to develop this scale: (1) define all relevant dimensions of AI technology/product readiness to be included in the scale; (2) explore the interrelationships among these dimensions; (3) develop a tool capable of accurately assessing AI technology/product readiness; and (4) provide a reference for different industries to determine their readiness to introduce AI technology/products.

This study addresses several critical gaps in the AI readiness literature. First, although prior research has underscored the importance of preparing for the large-scale deployment of AI systems, few methods currently exist to systematically monitor readiness levels and identify the key factors that facilitate this process (Pramanik et al., 2024). Second, most existing studies have predominantly focused on organizational or technological indicators of readiness (Cooper, 2025; Holmström, 2022; Jöhnk et al., 2021), while tools designed to effectively assess readiness at the individual level and reveal the factors influencing AI acceptance and adoption remain scarce—particularly within educational contexts (Li & Liang, 2025; Shahid et al., 2024). Consequently, the concept of "AI readiness" requires reconsideration and further conceptual refinement (Wang et al., 2023). To address these gaps, this study clarifies and advances the understanding of AI readiness by developing a multidimensional scale that encompasses four key dimensions: self-efficacy and cognition, optimism, collaborative learning and working, and comfort. This scale not only contributes to the theoretical advancement of AI readiness research but also provides practical insights for businesses, educators, and policymakers seeking to enhance preparedness for AI-driven innovation.

The remainder of this manuscript is structured as follows: Section 2 provides an overview of the essential theoretical foundations for formulating the dimensions of the AIRS. Section 3 presents the scale development process and data collection. Section 4 elaborates on the refinement outcomes of the AIRS during its developmental phase, including an assessment of the scale's measurement and dimensions of AI readiness. This section aims to establish the fundamental framework of the scale and conduct reliability assessments, while comprehensively evaluating its psychometric properties in terms of content, criterion-related, convergent, and discriminant validity. Section 5 extensively deliberates on the construction of the scale and explores its theoretical and practical implications. The concluding section discusses the results of applying the AIRS.

2. Theoretical foundations

2.1. Technology readiness and artificial intelligence

Technology readiness (TR), defined as the willingness of individuals to adopt and utilize new technologies to achieve personal or professional objectives, is shaped by psychological dispositions toward embracing technological advancements (Parasuraman, 2000). It has been widely applied across multiple domains and is commonly measured using the Technology Readiness Index (TRI), which is comprised of four primary dimensions: optimism, innovativeness, discomfort, and insecurity. Within this framework, optimism and innovativeness are viewed as positive drivers that facilitate technology acceptance and adoption, whereas discomfort and insecurity act as barriers that may hinder users' willingness to embrace new technologies (Wang & Wang, 2022). Specifically, optimism reflects positive perceptions of technology as enhancing control, efficiency, and quality of life (Cruz-Cárdenas et al., 2021; Parasuraman, 2000), while innovativeness represents a strong association with technology acceptance and the willingness to experiment with novel technologies (Ali et al., 2021). By contrast, discomfort stems from stress arising from a lack of control over technology (Walczuch et al., 2007), and insecurity reflects concerns about information security and privacy (Parasuraman, 2000; Son & Han, 2011).

Building on this theoretical foundation, prior studies have consistently highlighted TR as a key determinant of behavioral intention. Individuals with higher levels of TR are more likely to adopt technological innovations and use them more frequently (Han et al., 2024; Tang et al., 2021). TR has also been applied across diverse contexts to explain adoption behaviors. For instance, Wong et al. (2020) integrated TR into a framework for identifying determinants of blockchain adoption in supply chain management. D. Yin, Li, and Qiu (2023) investigated customer attitudes and behavioral intentions toward new technologies in AI-driven environments, while Flavián et al. (2022) examined the influence of TR on clients' intentions to adopt AI-based investment services. Collectively, these studies underscore the importance of TR as a critical construct for understanding both technology adoption and usage behaviors.

As artificial intelligence continues to be integrated across various domains, users' readiness has become an increasingly decisive factor in shaping the success of AI adoption (Alam, 2022; Haleem et al., 2019; Rajpurkar et al., 2022; Su et al., 2023; Zhai et al., 2021). For example, Kim et al. (2021) conducted an online experiment to compare individuals' attitudes toward social AI and functional AI, finding that participants perceived functional AI more positively due to its practical value. This highlights the need to understand users' readiness for AI technologies as a prerequisite for facilitating their widespread use (Jöhnk et al., 2021).

Further evidence of this relationship is provided by Issa et al. (2023), who examined the AI readiness of 464 agricultural technology professionals in the United States and Singapore. Their study explored how five technology stressors (e.g., overload, complexity, uncertainty) influence user engagement and experience. The findings reveal a positive correlation between stressors and engagement among U.S. participants, while certain stressors negatively affected usage behavior in the Singapore sample. This suggests that TR interacts with contextual factors, offering valuable insights into predicting and managing employees' responses to AI adoption in organizations and across cultures.

Nevertheless, AI applications continue to face challenges in practice. For instance, in the education sector, Malinka et al. (2023) observed that while AI assistants can enhance students' learning outcomes, the lack of effective plagiarism detection tools and enforcement mechanisms poses risks to academic integrity. Thus, beyond improving students' readiness to adopt AI, it is equally important to reassess educational models and ethical standards to mitigate potential misuse. This example illustrates that traditional

Table 1
Previous conceptualizations and factors of AI readiness.

Conceptualization	Usability Factors	Usability Evaluation Technique	Study
AIR refers to the assessments and preparations conducted by an organization during the process of adopting artificial intelligence.	Strategic alignment Resources Knowledge Culture Data	Qualitative interviews	Jöhnk et al. (2021)
AI readiness is an organization's ability to deploy and use AI in ways that add value to the organization.	Technologies Activities Boundaries Goals	Conceptual	Holmström (2022)
Medical artificial intelligence readiness refers to healthcare providers' preparedness in knowledge, skills, and attitudes to effectively use healthcare AI applications alongside their professional expertise in prevention, diagnosis, treatment, and rehabilitation.	Cognition Ability Vision Ethics	Survey	Karaca et al. (2021)
AI readiness is defined as the state of preparedness among teachers in terms of their cognition, ability, vision, and ethical considerations with respect to the use of AI in education.	Cognition Ability Vision Ethics	Survey	Wang et al. (2023)
AI readiness refers to the preparedness of individuals or organizations to adopt AI, considering factors like resources, strategic plans, and user perceptions. In education, it influences teachers' willingness to integrate AI into their teaching.	AI Teaching Knowledge AI Hardware Access AI Software Access AI Content Availability Administrative Support for AI Teaching	Survey	Jatileni et al. (2024)

TR indicators may not fully capture the nuances of individuals' attitudes and behaviors toward AI.

Empirical studies further confirm that TR significantly influences both the intention to adopt and the actual usage of AI technologies. For example, Aini et al. (2024) examined the adoption of an AI-based air quality monitoring solution that provides real-time assessments. Their results demonstrate that hedonic motivation and TR were major predictors of adoption intentions, suggesting that individuals with higher TR are more inclined to adopt and actively engage with AI-driven innovations.

In summary, the literature highlights the pivotal role of technology readiness in shaping the adoption and use of artificial intelligence. However, existing measures primarily focus on general technology acceptance and fail to adequately capture AI-specific characteristics. To address this gap, the present study builds upon the theoretical foundation of TR to develop a dedicated scale for AI readiness, systematically assessing individuals' psychological inclinations and adoption intentions toward AI technologies, thereby advancing both theoretical understanding and practical applications.

2.2. Artificial intelligence readiness

Artificial Intelligence Readiness (AIR) refers to the extent to which organizations or individuals are prepared to adopt and integrate AI technologies into their operations and daily activities. With the rapid advancement of AI technologies, AIR has emerged as a significant factor influencing technology adoption intentions. It has garnered widespread attention across various fields, including education (Chai et al., 2021), healthcare (Rony et al., 2024), and business (Issa et al., 2022). Table 1 summarizes the literature on the conceptualization and scale development of AIR, highlighting how researchers measure and analyze different aspects of this variable.

AIR is inherently multidimensional, encompassing individual and organizational preparedness in terms of knowledge, skills, resources, infrastructure, and managerial support necessary for the effective deployment of AI. At the individual level, Ziapour et al. (2025) examined medical students' readiness for AI in healthcare by identifying four dimensions: cognition, ability, vision, and ethics, underscoring the importance of adaptability in knowledge, skills, and attitudes. At the organizational level, Tehrani et al. (2024) conducted interviews with multinational enterprises and identified eight dimensions of organizational AI readiness: informational, environmental, infrastructural, participant, process, customer, data, and technological readiness—each representing critical factors required for successful AI implementation across different operational domains. Similarly, Yin et al. (2024) defined organizational AI readiness as the availability and preparedness of resources required for AI adoption, categorizing these into four major elements: financial resources, professional human resources, infrastructure, and top management support. Overall, a high level of AI readiness is considered essential for facilitating AI adoption and minimizing associated risks.

To further define the construct, Jöhnk et al. (2021) conducted interviews with 25 AI experts and identified 18 key factors of AIR, grouped into five core domains: strategic alignment (e.g., AI-business fit, customer AI readiness, executive support, AI-process alignment, and data-driven decision-making), resources (e.g., financial budgets, personnel, IT infrastructure), knowledge (e.g., AI cognition, skill enhancement, AI ethics), culture (e.g., innovation capacity, collaborative work, change management), and data (e.g., data availability, data quality, data accessibility, data flow). These domains constitute a comprehensive framework for assessing organizational AI readiness. Building on this, Holmström (2022) proposed an AIR assessment model to evaluate organizational capabilities for leveraging AI in digital transformation and to explore the associated values and challenges.

Beyond organizational contexts, several studies have explored individual-level AIR. For instance, Karaca et al. (2021) developed the Medical Artificial Intelligence Readiness Scale for Medical Students (MAIRS-MS), comprised of 22 items across four dimensions—cognition, ability, vision, and ethics. This scale assesses participants' understanding of AI-related terminology, ability to select appropriate AI tools, capacity to evaluate AI's advantages and limitations in healthcare, and adherence to AI-related legal and ethical standards. These findings align with Ziapour et al. (2025), reinforcing the significance of evaluating individual AI readiness.

Despite the growing interest in AIR, existing research remains fragmented, lacking a unified theoretical framework and focusing predominantly on organizational readiness, while paying comparatively little attention to the individual level (Cooper, 2025; Damerji & Salimi, 2021; Jöhnk et al., 2021). Yue et al. (2024) also emphasized that within the education sector, studies on AI readiness and attitudes toward AI-assisted teaching remain relatively scarce. To address this gap, the present study develops a measurement scale for assessing AI readiness at the individual level, thereby providing a more comprehensive basis for evaluation and fostering practical applications of AIR.

2.3. Artificial intelligence anxiety and readiness

With the rapid advancement and widespread application of AI and automation technologies, individuals are increasingly experiencing pressure from technological change. AI has reshaped industrial structures and labor markets, as well as altered individuals' perceptions of their future careers and social roles (Hui et al., 2024; Xue et al., 2022), thereby fueling widespread concerns about job displacement (Brynjolfsson & McAfee, 2014). Against this backdrop, Artificial Intelligence Anxiety (AIA) has emerged as a salient psychological construct, reflecting the emotional responses individuals exhibit toward AI technologies due to perceived threats and uncertainties. These responses may include fear of losing control, hesitation, and resistance to use (Constantinides et al., 2018; Johnson & Verdicchio, 2017; Wang & Wang, 2022). In this context, an individual's readiness—defined as the possession of sufficient AI-related knowledge, skills, and psychological adaptability—has been identified as a critical factor in coping with the pressures of technological transformation (Karaca et al., 2021).

Technological anxiety refers more broadly to the discomfort and fear elicited by emerging technologies, typically arising from a combination of unfamiliarity, negative attitudes, and perceived lack of control or competence (Brosnan, 1998; Ha et al., 2011). Wang and Wang (2022) emphasized that AIA is not only an individual emotional experience but also a collective phenomenon that reflects

societal transformation and cultural attitudes, potentially influencing the general pace and acceptance of AI adoption. As AI increasingly penetrates areas such as sales, marketing, programming, food service, and education, the risk of job displacement in these domains has become a major topic of public concern (Chounta et al., 2022; Nauman, 2017; Wang et al., 2023). It is important to note, however, that while the primary goal of AI implementation in organizations is often to enhance efficiency and support decision-making, rather than to completely replace human labor (Wilson & Daugherty, 2018), public anxiety surrounding the erosion of job opportunities due to automation remains prevalent. Willcocks (2020) argued that automation carries both the risk of substitution as well as the potential to create new types of jobs and redefine existing roles. Thus, AIA may stem more from anticipatory fear and information asymmetries than from actual threats.

In addition to employment-related anxiety, robot anxiety constitutes another important facet of AIA. Nomura et al. (2006) defined this concept as the aversive or fearful reactions individuals may experience when interacting with robots capable of human-machine interaction, which in turn can affect their willingness to accept and use such technologies. To assess this phenomenon, Wang and Wang (2022) developed the Artificial Intelligence Anxiety Scale, which encompasses four dimensions: learning, AI configuration, occupational allocation, and sociotechnical blindness in technology. This scale provides a comprehensive evaluation of individuals' emotional responses and attitudinal tendencies toward AI.

The occupational allocation dimension focuses on perceived impacts of AI on human behavior and values in the workplace (e.g., I am not worried that AI technology/products will make humans lazier), whereas AI configuration reflects instinctive emotional reactions toward humanoid AI or robots (e.g., I am not afraid of humanoid AI technology/products [i.e., robots]). These dimensions reveal the degree to which individuals are open to AI technologies as well as serve as predictors of their psychological readiness and behavioral tendencies in the face of digital transformation.

Artificial Intelligence Readiness (AIR), broadly defined as the degree to which individuals or organizations are willing and able to adopt and adapt to AI technologies in terms of knowledge, skills, and attitudes, has been shown to be negatively influenced by anxiety. Prior studies have demonstrated that anxiety significantly hinders technology acceptance behaviors and diminishes motivation for future learning and preparedness (Zhang & Lu, 2021). In other words, individuals who harbor fear and distrust toward AI are less likely to engage in learning about or applying such technologies, thereby creating a negative feedback loop.

In summary, AI anxiety undermines individuals' motivation to learn and accept AI technologies, thereby adversely affecting their AI readiness. If not effectively managed or reduced, AIA may impede the capacity of individuals and organizations to adapt to digital transformation. Consequently, understanding and addressing AI anxiety is of practical significance in terms of enhancing societal AI readiness.

2.4. Learning confidence and collaborative working

Developing well-defined concepts and establishing theoretical significance is critical for formulating reliable measures and producing valid outcomes (Gerbing & Anderson, 1988; Wang & Wang, 2022). In the context of AI Readiness, understanding both internal and external factors influencing readiness is essential. Internally, this involves self-efficacy—an individual's perceived ability to perform specific tasks (Bandura, 1997). For instance, technological self-efficacy, which refers to one's perceived ability to use technology to achieve goals, plays a key role in AI readiness (Compeau & Higgins, 1995; Hung et al., 2015; Sun & Rueda, 2012). The Mobile Learning Readiness (MLR) scale (Lin et al., 2016), widely applied to assess self-efficacy in mobile learning, exemplifies the relevance of domain-specific self-efficacy measures. The MLR scale contains 59 positive items (e.g., "I feel confident in my knowledge and skills related to mobile learning systems"). This focus on self-efficacy is directly applicable to AIR as it reflects an individual's preparedness to engage with AI technology.

Externally, perceived social support plays an equally important role in enhancing AI readiness. Social support refers to the help individuals receive from social networks, trusted friends, or family members when facing challenges (Lin et al., 2013; Shumaker & Brownell, 1984). The Multidimensional Scale of Perceived Social Support (MSPSS; Zimet et al., 1988), which measures support from family, friends, and significant others through 12 positive items (e.g., "There is a special person around me when I need them"), has demonstrated high reliability and can be adapted to assess external influences on AI adoption. The provision of social support can ease anxiety about new technologies and foster greater technological readiness, aligning with the AIR model's emphasis on preparedness through external interaction and assistance.

As AI technology becomes increasingly integrated into education and work, traditional modes of learning and collaboration are undergoing significant transformation (Rudolph et al., 2023). Tools such as ChatGPT are reshaping how individuals and teams engage with technology, fostering both learning confidence and collaboration (Hadi Mogavi et al., 2024; Javaid et al., 2023; Memarian & Doleck, 2023; Mhlanga, 2023; Pigg, 2024; Pinochet et al., 2023; Y. Su, Zhu, et al., 2023). Javaid et al. (2023) explored the transformative role of AI in Industry 4.0, finding that AI tools enhance collaboration between humans and machines, streamline manufacturing processes, and improve overall operational efficiency. Similarly, Mhlanga (2023) demonstrated that AI tools like ChatGPT can be leveraged in educational settings to promote student engagement, teamwork, and experiential learning, ultimately boosting learning outcomes.

These advancements highlight the critical connection between collaborative work and AI readiness. AI effectively enhances student engagement, teamwork, and learning, while also improving communication and decision-making skills (Rejeb et al., 2024). The AI readiness framework integrates personal self-efficacy and external social support to ensure that both individuals and organizations are adequately prepared for the adoption and utilization of AI technologies. Therefore, the successful integration of AI depends on individuals' learning confidence and collaborative structures that support effective technology adoption.

2.5. A theoretical framework for assessing AIR

AIR is expected to foster individuals' willingness to adopt AI technologies. Prior research on technology adoption has consistently shown that higher levels of readiness are associated with greater engagement in technology use (Flavián et al., 2022). For example, Chung et al. (2015) found that tourists with higher technological readiness were more likely to perceive augmented reality (AR) as useful, which in turn shaped positive attitudes toward AR and strengthened their intention to use AR in the future. Similarly, Issa et al.

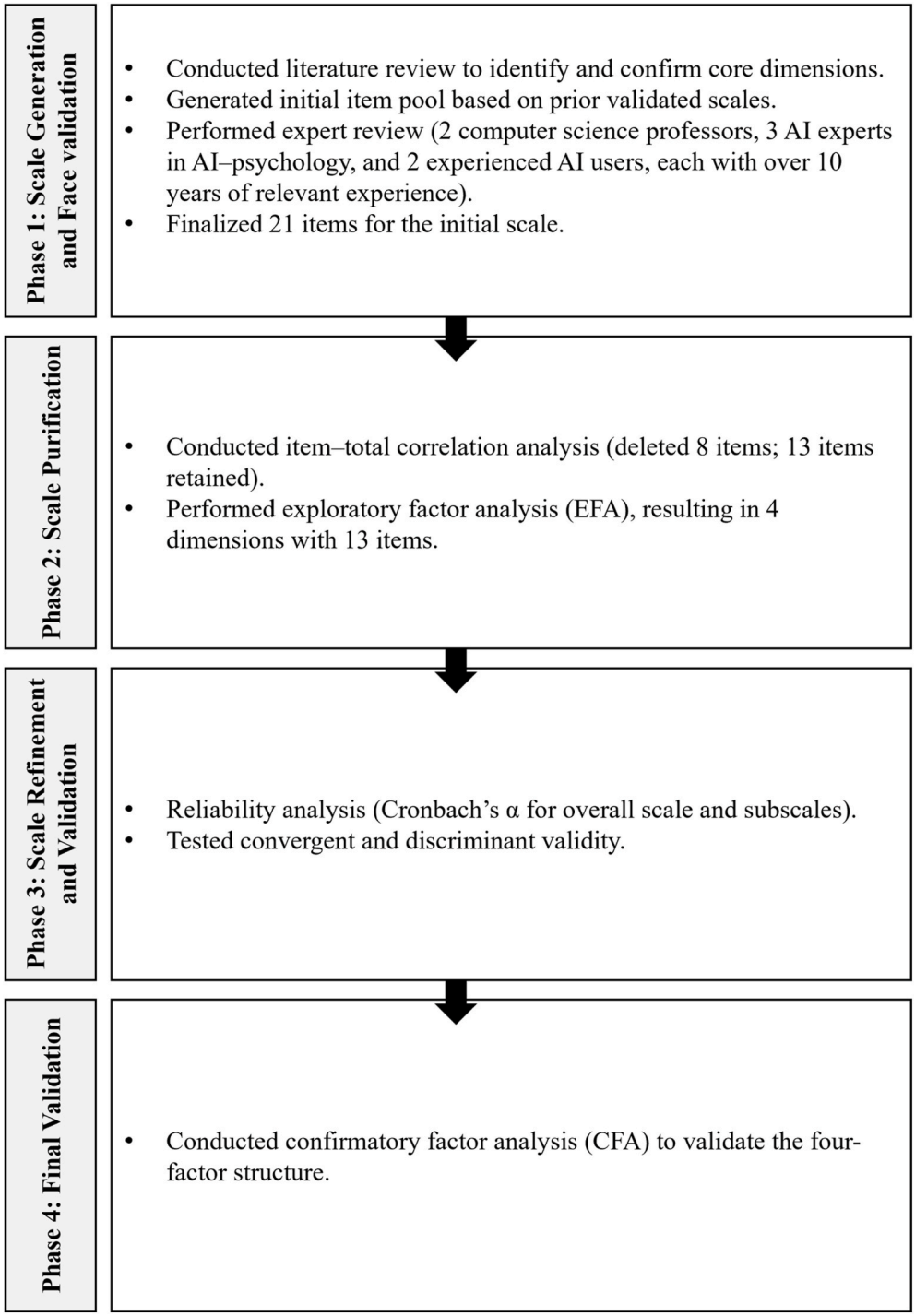


Fig. 1. Scale development procedure flowchart.

(2022) reported a positive relationship between AI readiness and AI adoption, while Baraka et al. (2025) highlighted that healthcare professionals' readiness for AI is strongly influenced by their intention to accept and use AI.

Building on these findings, this study posits that individuals with higher levels of AI readiness, reflected in greater self-efficacy, stronger perceived benefits, and higher comfort with AI, are more likely to adopt AI technologies. To establish the nomological validity of the AIR scale, we propose the following hypothesis.

H1. Individuals who have used AI technologies/products will report significantly higher levels of AI readiness than those who have not.

3. Methods

3.1. Scale development procedure

This study followed established guidelines and validation procedures for scale development (Carpenter, 2018; Hair et al., 2019; Lim et al., 2026) to ensure the rigor and validity of the AI Readiness scale. Its development followed a four-phase process. In Phase 1, a comprehensive literature review was conducted to identify and confirm the core dimensions of AI readiness. An initial item pool was generated based on prior validated scales and then reviewed by experts, including two computer science professors, three AI–psychology experts, and two experienced AI users with over 10 years of relevant experience. Based on their feedback, 21 items were finalized for the initial scale. In Phase 2, an item–total correlation analysis was performed, leading to the deletion of eight items and the retention of 13. An exploratory factor analysis (EFA) further refined the scale, yielding four dimensions across the 13 items. In Phase 3, a reliability analysis was conducted using Cronbach's α for both the overall scale and subscales, alongside tests of convergent and discriminant validity. Finally, a confirmatory factor analysis (CFA) was performed to validate the four-factor structure.

The overall process of scale development is illustrated in Fig. 1, which outlines the steps leading to the development and validation of the AIR scale.

3.2. Sample collection

This study employed a convenience sampling method, collecting data via an online questionnaire distributed through Google Forms. Participants were invited through social media and academic networks and were asked to select the response options that best described their AI-related experiences. Those interested in the study could access and complete the questionnaire via the provided link. Given that AI readiness may be influenced by various background factors, no restrictions were placed on participants' gender, age, or occupation, ensuring broader applicability of the findings. A total of 175 responses were collected, of which one was excluded due to invalid data, resulting in 174 valid responses. The demographic characteristics of the respondents are presented in Table 2.

Table 2
Demographic information of the respondents.

Characteristic	Item	Frequency	Percentage
Gender	Male	67	38.5
	Female	107	61.5
Age	15 or below	1	0.6
	16-20	8	4.6
	21-30	75	43.1
	31-40	34	19.5
	41-50	26	14.9
	51-60	26	14.9
	61 or above	4	2.3
Educational level	High school or below	18	10.3
	Bachelor's degree	114	65.5
	Master's degree	40	23.0
	Ph.D. degree or above	2	1.1
Occupation	Service	53	30.5
	Government	10	5.7
	Science and technology	12	6.9
	Education	11	6.3
	Manufacturing	13	7.5
	Student	29	16.7
	Medical	4	2.3
	Other	42	24.1
Have you ever used artificial intelligence technologies/products	Yes	153	87.9
	No	21	12.1

4. Development of the AIRS

4.1. Dimension identification and item generation

This study conducted a comprehensive literature review to identify the core constructs and relevant measurement dimensions of AI readiness. Through the review and analysis of related literature, we identified the key dimensions of AI readiness, including self-efficacy, anxiety, social support, collaborative learning and work (CLW), optimism, innovativeness, and discomfort. The fundamental dimensions were summarized into a dimension table (see Table 3).

Based on the identified constructs, we further analyzed existing validated scales to inform the design of our measurement items. For example, self-efficacy items were adapted from the MLR scale (Lin et al., 2016); anxiety items were based on the AI Anxiety Scale (AIAS; Wang & Wang, 2022); social support items referred to the Multidimensional Scale of Perceived Social Support (MSPSS; Zimet et al., 1988); and the CLW dimension was informed by Javaid et al. (2023) and Mhlanga (2023). Additionally, the Technology Readiness Index (TRI; Parasuraman, 2000) served as a theoretical framework to conceptualize optimism, innovativeness, and discomfort, thereby enriching the content of the AIR scale. A preliminary pool of items was developed based on these sources (see Appendix A). During item construction, we carefully reviewed the wording, phrasing, and underlying meanings of existing items to ensure the newly developed items aligned with the language and experiential expressions of the target respondents. This process enhanced the relevance and applicability of the scale (Khatri et al., 2025).

Following the initial item development, an expert validation procedure was conducted to ensure the face validity of the scale. The expert panel included two computer science professors with over 10 years of teaching and research experience, three AI experts with over a decade of interdisciplinary research experience in AI and psychology, and two experienced AI users with more than 10 years of experience in AI technologies or products. These experts were invited to assess the appropriateness and comprehensiveness of all items. Items unanimously approved by the expert panel were retained in the formal version of the scale.

After expert evaluation and revision, a total of 21 initial items measuring the AIR construct were finalized, with appropriate modifications made based on expert feedback (e.g., refinement of vocabulary and phrasing). The finalized AIR scale consisted of 21 items rated on a five-point Likert scale, ranging from “strongly disagree” (1) to “strongly agree” (5). Higher (or lower) scores indicated a higher (or lower) evaluation of the corresponding dimensions and items.

4.2. Scale purification

The initial scale was comprised of 21 items, excluding demographic variables and items used in scale validation. Cronbach's α and item-total correlations were used to identify and delete inappropriate items. Moreover, to mitigate false part-whole correlations (Cohen & Cohen, 1975), corrected item-to-total correlation was also employed as a criterion for item deletion. Consequently, the Cronbach's α for the developed scale was set at >0.5 ($\alpha > 0.7$ is preferred; Nunnally, 1978), with higher values indicating greater reliability. Standard practice in earlier studies was to eliminate items on the initial scale with item-total correlation coefficients <0.4 (Cristobal et al., 2007; Cronbach, 1951; Everitt, 2002). The above criteria were used to identify and remove items that lacked discriminant validity. The Cronbach's α of the initial scale was 0.875. Of the initial items, the 13 with item-total correlations greater

Table 3
Dimension table of AI readiness.

Dimension	Javaid et al. (2023)	Jöhnk et al. (2021)	Karaca et al. (2021); Wang et al. (2023)	Li and Liang (2025)	Lin et al. (2016)	Mhlanga (2023)	Parasuraman (2000)	Wang and Li (2024)	Wang and Wang (2022)	D. Yin, Li, and Qiu (2023)
Self-efficacy					✓			✓		✓
Awareness		✓								✓
Perceived value and cost				✓						
Facilitating condition				✓						
Cognition			✓							
Ability		✓	✓					✓		
Vision			✓					✓		
Ethics		✓	✓							
Anxiety					✓				✓	
Social support		✓		✓				✓		
Optimism					✓		✓	✓		✓
Collaborative learning and working	✓	✓				✓				
Innovativeness		✓		✓			✓	✓		✓
Discomfort/ Comfort							✓	✓		
Insecurity		✓					✓			

than 0.4 were retained, while the 8 with item-total correlations less than 0.4, including Q4, Q5, Q6, Q7, Q10, Q11, Q15, and Q17, were omitted.

Further validation of the basic structure of the resulting 13-item scale was conducted using an EFA, which can reveal the latent structures among variables to facilitate a deeper understanding of these variables and their interrelationships. Before using the EFA to identify the structural elements involved in AIR, the Kaiser–Meyer–Olkin (KMO) measure and Bartlett's test of sphericity were employed to ensure the appropriateness of the data for factor analysis (Hair, Black, Babin, Anderson, & Tatham, 2010). The adequate score earned on the KMO measure (MSA >0.5) and significant result achieved on the Bartlett's test ($\chi^2 = 1484.668$; $p < 0.001$) confirmed common factors in the population correlation matrix, implying the suitability of using factor analysis. Furthermore, a principal component analysis (PCA) and an orthogonal method (varimax) were used to explore and identify the key underlying factors affecting preparedness to accept AI among the sample. The factor eigenvalues in the analysis results were greater than 1, accounting for 76.564 % of the total extracted percentage variance in the AIRS. In employing EFA to assess the single-dimensional convergence and discriminant validity of the AIRS, the following decision-making criteria for item deletion were used (Hair et al., 2010; Hair et al., 1998; Straub, 1989): (1) extracted factor eigenvalues >1; (2) factor loadings <0.5; and (3) items cross-loaded across factors.

Based on the above rules and EFA results, four factors (dimensions) with 13 items were established for subsequent analysis and coefficient α assessment. These four dimensions included self-efficacy and cognition (SEC), optimism, CLW, and comfort. The factor loadings of these 13 items are shown in Table 4, with each item showing convergent validity and loading on a single dimension. No cross-loaded items were found in any of the four factors, providing sufficient support for the discriminant validity of the AIRS.

4.3. Reliability test

Reliability analysis was performed again to ensure the reliability of the developed scale. The Cronbach's α coefficient was 0.92 for the 13 items of the AIRS, which is significantly greater than the minimum recommended value of 0.7 (Hair et al., 2010). The individual reliabilities of the four factors were as follows: self-efficacy and cognition = 0.89, optimism = 0.90, CLW = 0.87, and comfort = 0.72. Furthermore, as shown in Table 5, the corrected item-total correlation coefficients for all 13 items exceeded the minimum value of 0.40 (Everitt, 2002; Field, 2005). Thus, the reliability of the AIRS was confirmed based on the analysis results.

4.4. Content validity

The AIRS demonstrated robust internal consistency and a straightforward, easy-to-understand, and stable factor structure. The high internal consistency of this scale, a critical factor used to evaluate construct validity, indicates that each item effectively measures its associated construct and that uniformity among the test items is high. Content validity is as a qualitative criterion used to determine whether measurement items sufficiently encompass the comprehensiveness, representativeness, and appropriateness of the measured concept and thus fully capture the overall theoretical concept (Carmines & Zeller, 1979; Haynes et al., 1995). Content validity may be assessed by using scale items to measure attributes, inviting relevant experts or scholars to assess the appropriateness of scale items, or other means (Churchill, 1979; Dorussen et al., 2005; Rubio et al., 2003; Yaghmaie, 2003). Lim (2024) emphasized that engaging domain experts to evaluate whether measurement items adequately capture the intended construct and continuously refining these items until no additional modifications are suggested is an important strategy for enhancing content validity. In line with this approach, the 13 items of the AIRS scale in this study were carefully selected based on relevant literature and then reviewed by a panel of experts with extensive professional backgrounds. The expert panel included two researchers, two AI users with experience, and three AI experts. They assessed each item in terms of its appropriateness, completeness, and relevance to the theoretical construct. After comprehensive discussions, only items that received unanimous approval from all experts were retained in the final scale. This rigorous process ensured that each item accurately and comprehensively reflects the intended theoretical construct, thereby establishing a high level of content validity for the scale.

Table 4
Rotated factor loadings for the 13-item AIRS.

Item code	SEC	OPT	CLW	COM
Q13	0.841			
Q12	0.790			
Q14	0.770			
Q16	0.703			
Q21	0.601			
Q1		0.897		
Q2		0.843		
Q3		0.770		
Q20			0.802	
Q19			0.775	
Q9				0.879
Q8				0.719
Q18				0.516

Notes: SEC: self-efficacy and cognition; OPT: optimism; CLW: collaborative learning and working; COM: comfort; Values below 0.50 were omitted.

Table 5
Item-to-total correlations for AIR measures.

Item code	Original item code	Item description	Corrected item-to-total correlation
SEC1	Q13	I have confidence in learning to operate AI technologies/products.	0.729
SEC2	Q12	I have confidence in understanding how AI technologies/products operate.	0.737
SEC3	Q14	I have confidence in utilizing the various functions provided by AI technologies/products.	0.757
SEC4	Q16	I do not feel anxious about learning to use AI technologies/products.	0.649
SEC5	Q21	I believe that AI technologies/products can understand the content of collaborative work.	0.708
OPT1	Q1	I believe that AI technologies/products enable me to better manage daily life.	0.636
OPT2	Q2	I believe that AI technologies/products make my daily life more efficient.	0.639
OPT3	Q3	I like AI technologies/products because I can use them anytime.	0.709
CLW1	Q20	I believe that collaborating with AI technologies/products can improve work efficiency.	0.705
CLW2	Q19	I believe that AI technologies/products are very useful tools for education and learning.	0.679
COM1	Q9	I do not feel embarrassed when encountering problems or difficulties while using AI technologies/products in front of others.	0.459
COM2	Q8	I believe many artificial intelligence technologies/products do not raise health or safety concerns.	0.632
COM3	Q18	I am not afraid of humanoid AI technologies/products (robots).	0.613

Notes: SEC: self-efficacy and cognition; OPT: optimism; CLW: collaborative learning and working; COM: comfort.

4.5. Criterion-related validity

Criterion-related validity refers to the closeness of the relationship between the characteristics of a measurement tool and an external criterion (Houser, 2012). Thus, a scale designed to measure the AIR construct should exhibit a positive correlation between the total scale score and the valid criterion. In this study, a criterion-related item asking whether the respondent has ever used AI products/technology was designed to assess criterion-related validity. The correlation analysis between the criterion-related item and the 13-item AIRS yielded a criterion-related validity of 0.254 with a significance level of 0.001, demonstrating acceptable criterion-related validity (Cohen, 1988).

4.6. Convergent and discriminant validity

Convergent validity was assessed following the approach of Kaiser (1958) in terms of ensuring that (1) extracted factor eigenvalues exceed 1, (2) factor loadings of individual items within each factor exceed 0.5, and (3) cumulative explained variance exceeds 50%. The results, presented in Table 4 and described in Section 4.2, indicate the AIRS meets the criteria for convergent validity.

Discriminant validity was assessed using the criteria suggested by Gaski and Nevin (1985), which include ensuring that (1) the correlation coefficients between any two variables are less than one and (2) the correlation coefficients between any two constructs are less than the individual constructs' Cronbach's α coefficients. As shown in Table 6 and described in Section 4.3, the correlation coefficients between the constructs fulfill the abovementioned criteria.

4.7. Nomological validity

To examine the nomological validity of the AIRS, we tested the relationship between the 13-item AIR scale and actual AI usage behavior. Nomological validity helps verify how well the construct fits within a theoretically supported network of related variables (Lim, 2024). Prior research has shown that AI readiness is strongly influenced by individuals' willingness to accept and use AI technologies (Baraka et al., 2025). Accordingly, individuals with prior experience using AI products are expected to demonstrate higher levels of AI readiness. In this study, nomological validity was evaluated through H1, which tested the association between the AIRS and a theoretically relevant variable, namely AI usage. Participants were divided into two groups based on whether they had previously used AI technologies/products, and their overall AIRS mean scores were compared. A significantly higher AIRS score among users, compared with non-users, would provide support for H1 and confirm the nomological validity of the scale.

The results revealed a significant difference in AI readiness between participants with and without prior AI experience ($t = 3.45$, $p < 0.001$). Specifically, participants who had used AI technologies/products reported significantly higher AIR scores ($M = 3.94$, $SD =$

Table 6
Correlations among AI readiness dimensions.

	Factor 1: SEC	Factor 2: OPT	Factor 3: CLW	Factor 4: COM
Factor 1	1			
Factor 2	0.586**	1		
Factor 3	0.664**	0.618**	1	
Factor 4	0.647**	0.492**	0.503**	1

Notes: SEC: self-efficacy and cognition; OPT: optimism; CLW: collaborative learning and working; COM: comfort; values below 0.50 were omitted.
**$p < 0.01$.

0.64) than those without such experience ($M = 3.44$, $SD = 0.51$). Thus, H1 was supported, providing empirical evidence for the nomological validity of the proposed AIR measurement.

4.8. Norm of the AIR scale

After confirming the reliability and validity of the developed scale, norms were established (MacKenzie et al., 2011) to determine respondent distributions on specific constructs. Norms allow practitioners to assess the position of individual scale respondents relative to others (Lin et al., 2016; Wang, 2007), improving their ability to apply strategic thinking to related problems. Although the AIRS may be used to assess an individual's AIR, comparing individual readiness against the standard to the total distribution of readiness is an even better method of assessment.

As shown in Table 7, an AIRS total score of 49 or less ranks in the bottom-half of the total sample, indicating poor AIR. Conversely, a total score of 50 or more indicates better AIR and likely more optimism, learning confidence, and collaborative cooperation in the context of technology. For other relevant data, with an average of 50.43, a median of 49, a minimum value of 29, a maximum value of 65, a standard deviation of 8.349, skewness of 0.247, and kurtosis of −0.226. In general, data with skewness and kurtosis values in the range of +2.58 to −2.58 are considered to have normality at a confidence level of less than 99%. The skewness and kurtosis values in this study were 0.247 and −0.226, respectively, indicating that the collected data and the AIRS follow a normal distribution.

4.9. Confirmatory factor analysis

This study employed a Confirmatory Factor Analysis (CFA) to further validate the model established through EFA. CFA is primarily used to examine the degree of fit between a theoretical model and the observed data, thereby assessing whether each measurement item effectively reflects its underlying latent construct (Henson & Roberts, 2006). Model fit was evaluated using several indices, including the chi-square test (χ^2), the chi-square to degrees of freedom ratio (χ^2/df), the Comparative Fit Index (CFI), the Non-Normed Fit Index (NNFI/TLI), the Root Mean Square Error of Approximation (RMSEA), the Standardized Root Mean Square Residual (SRMR), and the Goodness-of-Fit Index (GFI). Generally, acceptable model fit is indicated when χ^2/df is less than 3, CFI and NNFI exceed 0.90, RMSEA is below 0.1, SRMR is below 0.08, and GFI exceeds 0.90 (Bentler, 1982; Hair, Black, et al., 2019; Hu & Bentler, 1999; MacCallum et al., 1996; McDonald & Ho, 2002).

This study compared the fit of a one-factor model with that of a four-factor model. The one-factor model demonstrated inadequate fit indices (e.g., CFI = 0.759, NNFI = 0.711, RMSEA = 0.176, SRMR = 0.084, GFI = 0.709), suggesting that the convergence of all items on a single factor is not appropriate. In contrast, the four-factor model showed substantially better fit (e.g., χ^2/df = 2.31, CFI = 0.947, NNFI = 0.93, SRMR = 0.048, GFI = 0.9, RMSEA = 0.087), with values that meet conventional thresholds for acceptable model fit. Moreover, the chi-square value of the four-factor model ($\chi^2 = 136.139$) was significantly lower than that of the one-factor model ($\chi^2 = 417.018$), further supporting the superiority of the four-factor structure. Taken together, these results validate the four-factor model as a robust theoretical foundation for analyzing AI Readiness, demonstrating that each item effectively reflects its corresponding construct.

5. Discussion

Based on previous scale development standards (norming), this study employed factor analysis, reliability analysis, discriminant validity analysis, and similar methods to establish the latent factors of artificial intelligence readiness. Subsequently, the Artificial Intelligence Readiness Scale (AIRS), which is a reliable and comprehensive multidimensional scale, was developed. Since the emergence of generative pretrained transformers (GPT), artificial intelligence applications have gradually become more integrated into everyday life and familiar to the general public. From virtual digital assistants on smartphones to medical diagnostic systems, and now prevalent applications in fields such as financial technology and educational learning, AI is ubiquitous. Despite significant public interest in AI and its future implications, its widespread impact can only be realized through technological transformation. This transformation occurs when new ideas and specialized knowledge mature and become widely applicable (Martínez-Plumed et al.,

Table 7
13-item AI readiness scale: Percentile scores.

Percentile	Value
10	41
20	45
25	46
30	46
40	47
50	49
60	51
70	53
75	55
80	58
90	65

2021). As AI technologies/products increasingly play pivotal roles in enterprises and become more practical in daily environments, research on whether people are prepared for AI to usher in significant life changes remains scarce.

Although the technology readiness level (TRL) has previously been used to assess AI technologies, and a close relationship between TRL and AI technologies has been found (Martínez-Plumed et al., 2021), no validated scale existed to evaluate readiness for AI. Thus, the development of a framework for assessing AIR by referencing the literature related to technology readiness is the main theoretical contribution of this work. In this study, a generic AIRS with satisfactory psychometric properties was rigorously developed and tested. Empirical testing was used to establish the 13-item AIRS encompassing the four main factors of learning confidence, collaborative working, optimism, and comfort. The analyses conducted demonstrate the AIRS has acceptable reliability as well as expertise, content, criterion-related, convergent, and discriminant validities. The approach used by the model to measure the AIR construct is illustrated in Fig. 2.

The AIR concept was developed in this study by integrating technology readiness, AIA, and self-efficacy, as well as referencing psychological dimensions, e.g., CLW, from the literature. Notably, the developed AIRS comprises the four dimensions of (1) self-efficacy and cognition (the learning dimension of the AIA scale integrated CLW perspectives), (2) optimism (an aspect of technology readiness-related optimism), (3) learning confidence and collaborative working (CLW; a unique aspect of AIR), and (4) comfort (a dimension of maladaptation from the TRI that includes aspects of AI configuration from the AIA scale). Consistent with prior research on AIA (Wang & Wang, 2022), a relationship between anxiety and AIR was identified in this study, suggesting that reducing users' perceived anxiety is key to accepting AI technology. Related strategies may facilitate more extensive use of AI products and collaborative learning methods, aligning with the CLW dimension of the AIRS. In addition, the relationship identified between AIR and the optimism and comfort dimensions support the validity of applying technology readiness factors to AI (Parasuraman, 2000). The primary factors comprising AIR proposed in this study may be applied to AI technology/product adoption and development in various industries. Moreover, the different aspects in the AIRS may be used to help individuals assess their readiness to accept AI technologies and products.

The 13-item AIRS demonstrated acceptable reliability and criterion-related content as well as convergent and discriminant validity. Based on the analysis results given in Table 8, scores for the self-efficacy and cognition (SEC) dimension below 19 (i.e., 50th percentile) are in the bottom half of the sample. Self-efficacy refers to the beliefs and perceptions of an individual in their own abilities and is the most important determinant of behavioral change (Bandura, 1997). Higher confidence in this dimension may be expected to enhance AIR. With regard to the optimism dimension, scores below 12 (i.e., 50th percentile) are in the bottom half of the sample. As in previous research, which identified optimism as a factor that motivates individuals to adopt technologies/products (Cruz-Cárdenas et al., 2021), this study found a positive relationship between optimism and AIR. With regard to the CLW dimension, scores below 8 (i.e., the 50th percentile) are in the bottom half of the sample. It is critical for individuals to learn the workings, capabilities, and limitations of AI to use it confidently and positively (Alam, 2022). Educators can foster deeper and more responsible connections between individuals and

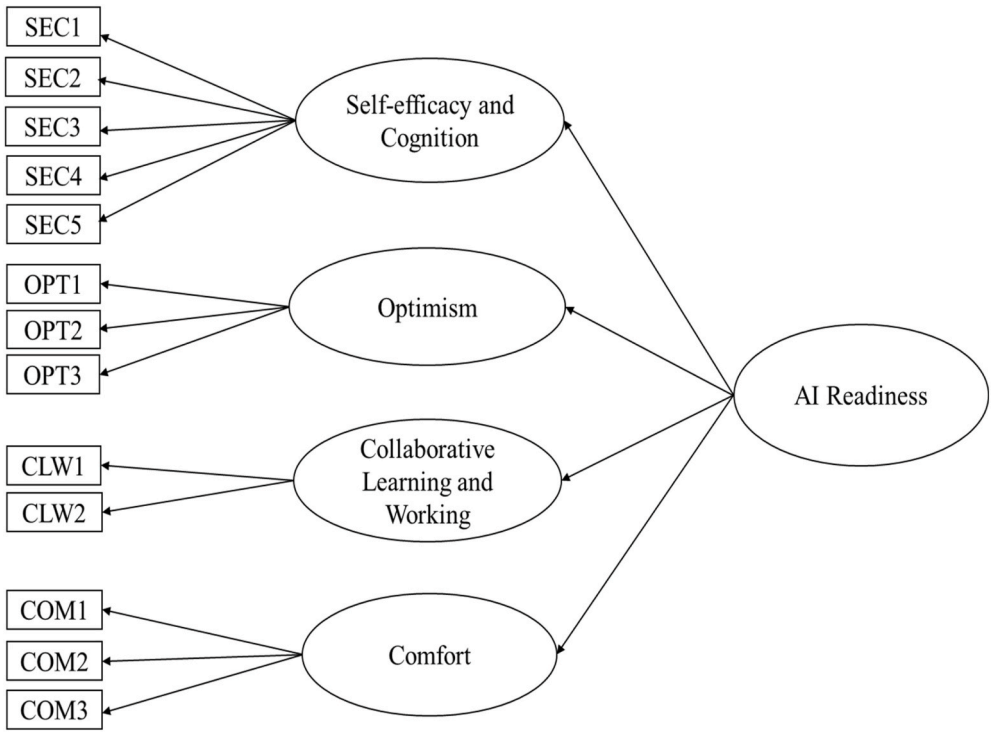


Fig. 2. A model for measuring artificial intelligence readiness.

Table 8
Percentiles of each AIRS dimension.

Percentile	Factor Score			
	SEC	OPT	CLW	COM
10	15	10	7	7
20	16	11	8	8
25	17	11	8	8
30	17	12	8	8
40	19	12	8	9
50	19	12	8	9
60	20	13	9	10
70	21	14	10	11
75	22	14.25	10	11
80	25	15	10	12
90	25	15	10	15

Notes: SEC: self-efficacy and cognition; OPT: optimism; CLW: collaborative learning and working; COM: comfort.

AI by teaching about its limitations, thus enabling more people to understand its scope and capabilities (Mhlanga, 2023) and cultivating the confidence necessary to leverage AI in both their academic pursuits and subsequent careers. With regard to the comfort dimension, scores below 9 (i.e., the 50th percentile) are in the bottom half of the sample. These findings suggest that encountering problems when using AI in public is not a source of embarrassment, contradicting the idea that people are hesitant to try or use unfamiliar/new technologies in public (Cruz-Cárdenas et al., 2021). Building on the widespread functional knowledge related to AI and AI applications, the findings of this study indicate that recent technological advances have extended the role of AI into the realms of social and interpersonal relationships (Kim et al., 2021), which exhibit a lower AIR threshold. Therefore, artificial intelligence and humanoid artificial intelligence are like friends, and even if problems occur, they will not have a negative impact.

5.1. Theoretical implications

This study makes significant theoretical contributions to the literature on AI readiness. First, it develops and validates the AIRS, offering a reliable and multidimensional framework for assessing individuals' adaptability to AI technologies. While previous studies have primarily focused on the adoption of AI technologies, there remains a lack of systematic exploration of the factors influencing AI readiness (Jöhnk et al., 2021). By establishing a standardized measurement tool, this study addresses that gap and enhances the generalizability of AI readiness assessments.

Second, this study extends the application of the Technology Readiness Index (TRI) by validating the influence of self-efficacy, optimism, collaborative learning, and comfort on AI readiness. These dimensions enrich our understanding of individual-level AI adoption and complement existing technology readiness theories. By incorporating these factors, the research provides a more holistic view of how individuals perceive and adapt to AI technologies.

Third, the study emphasizes the critical role of psychological and cognitive preparedness in AI adoption, shedding light on the emotional and cognitive processes that underpin individual adaptation. As noted by Uren and Edwards (2023), AI-driven innovation depends not only on technological advancements but also on comprehensive user preparedness. The AIRS developed in this research offers a theoretical foundation for future studies to explore the implications of AI readiness at individual levels, contributing to broader discussions on AI adoption and its societal impact.

5.2. Practical implications

The findings of this study hold substantial practical implications for multiple stakeholders, including policymakers, organizations, educators, and AI developers. For organizations, AIRS serves as a valuable diagnostic tool to assess employees' readiness for AI-driven workplace transformations. By identifying gaps in AI readiness, organizations can design targeted training programs to enhance workforce adaptability, reduce anxiety surrounding AI implementation, and foster a culture of collaborative learning. As Uren and Edwards (2023) highlighted, cultivating AI-related skills can help alleviate employee concerns about technological transitions.

For educators, AI readiness is a critical factor in the successful integration of AI-enhanced education (Wang et al., 2023). Teachers' awareness and competence regarding AI directly influence students' acceptance and use of such technologies. The AIRS provides a foundation for exploring individuals' AI-related self-efficacy and psychological factors, supporting the effective adoption of AI in educational settings. Understanding the current state and needs of educators' AI readiness can inform the development of tailored training initiatives and strategic resource allocation to ensure a smooth integration of AI into teaching practices.

For policymakers, the study offers insights into national AI readiness and digital inclusion strategies. Governments can use AIRS to evaluate the public's preparedness across psychological, knowledge-based, and skills dimensions. This facilitates the creation of inclusive policies targeting various demographic groups, such as older adults, individuals with lower educational attainment, or digitally marginalized populations. These efforts promote equitable access to AI technologies and support nationwide initiatives aimed at improving AI literacy, designing educational programs, and allocating regional resources. For example, Chohan and Hu (2022) found that e-government training significantly enhances users' self-efficacy in utilizing public services, helping to bridge the digital divide

and promote equitable access. Additionally, the regular administration of AIRS can enable policymakers to monitor shifts in national AI readiness over time, offering an empirical foundation for long-term digital transformation planning and policy evaluation, ultimately reducing digital inequality and enhancing technological competitiveness.

For AI developers, understanding users' AI readiness is crucial to designing inclusive and user-centered technologies. AIRS can serve as a preliminary diagnostic tool for identifying psychological, knowledge, and skill-related differences across user groups. These insights can guide the development of user interfaces and interaction flows that align with user needs. For instance, users with lower levels of AI readiness may benefit from systems that offer more guidance, contextual prompts, and real-time feedback to reduce cognitive load. In contrast, users with higher readiness may prefer advanced features and greater operational flexibility to enhance efficiency and satisfaction.

As AI continues to evolve, ensuring individuals across sectors are adequately prepared is essential for maximizing the benefits of AI in both professional and personal domains. Future research and policy initiatives can build upon these findings to promote a more inclusive approach to AI integration.

6. Conclusion and limitations

The integration of artificial intelligence into everyday life is progressing as cloud computing transitions to edge computing and machine learning advances to generative AI. Although AI research is expanding in scope and diversity, research on measuring Artificial Intelligence Readiness (AIR) is lacking. This study aims to address this gap by introducing and validating the Artificial Intelligence Readiness Scale (AIRS) as a standardized tool to evaluate the readiness of the general public for AI. This study investigated the interconnections among various factors influencing AI readiness, such as self-efficacy, cognition, optimism, collaborative learning and working, and comfort. The development of the AIRS marks a significant advancement in the conceptualization and assessment of AI readiness. By establishing a theoretical framework, designing operational procedures for a set of initial AIR items, and empirically validating the comprehensive AIRS, we confirmed the robust psychometric properties of the proposed scale. This validation will be beneficial for AI developers and practitioners in the deployment of AI technologies and products. Additionally, educators and researchers can utilize the AIRS to develop and test theories related to AI adoption readiness. This study also offers preliminary insights into the correlation between AI readiness, collaborative learning, and work.

6.1. Limitations and future research

Despite the comprehensive development of the AIRS, this study has several limitations that warrant consideration. These limitations are delineated below along with suggestions for future research endeavors in this domain. First, the broad spectrum of applications and presentation formats associated with AI products was not explicitly defined within the study, necessitating further investigation to determine the generalizability and reliability of the AIRS across diverse categories of AI products.

Second, future research should focus on refining the 13-item structure to further assess its psychometric robustness. In particular, test-retest reliability is needed to evaluate the stability of the instrument over time. Common methods for estimating reliability include internal consistency reliability, which is typically measured using indices such as stability, reliability, and Cronbach's alpha, while test-retest reliability evaluates the instrument's consistency over time. The test-retest approach is particularly valuable for establishing a dependable scale (Galletta & Lederer, 1989). Therefore, further investigations utilizing the test-retest methodology are warranted to evaluate the reliability of the AIRS in terms of both short- and long-term stability.

Finally, all of the study participants were from Taiwan, their ages ranged from 20 to 40 years, and prior experience using AI products was not assessed. Factors such as geographical region, age group, and prior experience with AI products may influence values and thinking patterns and thus affect sample biases. Subsequent research should address these limitations to make this study more representative and broadly applicable.

CRediT authorship contribution statement

I-Ying Wang: Writing – original draft, Methodology, Investigation, Conceptualization. **Cheng-Yu Hung:** Writing – original draft, Methodology, Investigation, Conceptualization. **Hsin-Hui Lin:** Writing – review & editing, Validation, Methodology, Investigation, Conceptualization. **Wen-Hsien Lai:** Software, Methodology, Formal analysis, Data curation, Conceptualization. **Yi-Shun Wang:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Competing interest declaration

The authors have no conflict of interest to declare.

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Appendix A. Items used to measure AI readiness in this study

1. I believe that AI technologies/products enable me to better manage daily life.
2. I believe that AI technologies/products make my daily life more efficient.
3. I like AI technologies/products because I can use them anytime.
4. I think my friends know more about AI technologies/products than I do.
5. I find that I encounter fewer challenges with AI technologies/products compared to others.
6. Generally, I am among the first of my friends to acquire AI technologies/products when they are released.
7. I believe AI technologies/products are not designed for ordinary people.
8. I believe many artificial intelligence technologies/products do not raise health or safety concerns.
9. I do not feel embarrassed when encountering problems or difficulties while using AI technologies/products in front of others.
10. When I have questions about AI technologies/products, my friends help me.
11. Whenever I have questions about AI technologies/products, I can always find someone who can help me.
12. I have confidence in understanding how AI technologies/products operate.
13. I have confidence in learning to operate AI technologies/products.
14. I have confidence in utilizing the various functions provided by AI technologies/products.
15. I am not worried that AI technologies/products will make humans lazier.
16. I do not feel anxious about learning to use AI technologies/products.
17. I am not worried that AI technologies/products will be abused.
18. I am not afraid of humanoid AI technologies/products (robots).
19. I believe that AI technologies/products are very useful tools for education and learning.
20. I believe that collaborating with AI technologies/products can improve work efficiency.
21. I believe that AI technologies/products can understand the content of collaborative work.

Data availability

Data will be made available on request.

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